TechSol AI Engineer Assignment — Build a BA Meeting Agent

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Build an **AI Business Analyst agent** that joins a TechSol car-design meeting, answers when called, takes notes with who said what and when, and writes tasks to Notion.

What to build

Area	Requirement	Must
Meeting	Joins Google Meet, Zoom or Teams (your choice) as "TechSol BA"; replies in chat within 10 s to "Hey BA …"; silent otherwise	✓
Notes	Decisions, tasks, questions and risks — every line tagged with speaker and timestamp	✓
Notion	Fills 4 linked databases: Meeting Notes, Tasks, Decisions, Requirements; tasks have owner, due date, status; no duplicates	✓
Knowledge	Car engineering (EV, chassis, NCAP, ADAS, regulation, BOM) and BA outputs (user stories, MoSCoW); says "not sure" instead of inventing	✓
Car parts	Knows main parts by system (battery pack, motor, inverter, chassis, brakes, suspension, body, ADAS sensors, infotainment); builds a BOM table with part, qty, estimated cost and supplier from our parts pack; flags single-supplier and long-lead parts	✓
Memory	Recalls earlier points in the same meeting	✓
Deploy	`docker compose up`, deployed on our cloud VM; LLM swappable by config, self-hosted by default	✓
Bonus	Voice replies, second platform, cross-meeting memory, Risks database	+10

The AI agent

TechSol BA acts like a senior business analyst with automotive engineering background: it listens quietly, keeps the record, and speaks only when asked — short, factual, with assumptions stated.

Part	What we expect	Must
Role & persona	System prompt defines it as TechSol's BA: neutral, concise, never takes sides, asks for owner/date when missing	✓
Listener	Streams meeting audio to speech-to-text, keeps a rolling transcript with speaker + time	✓
Trigger	Detects "Hey BA" and extracts the question; ignores everything else	✓
Classifier	Tags each transcript chunk as Decision / Task / Question / Risk / Requirement / Note	✓
Reasoning (LLM)	Plans the answer, decides which tool to call, writes the reply	✓
Knowledge (RAG)	Searches the automotive pack (specs, parts, BOM, suppliers) before answering; cites the source doc	✓
Short-term memory	Full current-meeting context: decisions, tasks, earlier answers	✓
Long-term memory	Past meetings from Notion / vector DB	Bonus
Guardrails	Refuses off-topic or unsafe requests; says "not sure" when the pack has no answer	✓

Tools the agent must call

Tool	Does	Must
`post_chat(text)`	Writes a reply in the meeting chat	✓
`search_knowledge(query)`	Retrieves from the automotive / parts pack	✓

Tool	Does	Must
`create_task(task, owner, due, said_by, time)`	Adds a row to Notion Tasks, skips duplicates	✓
`update_task(task, status)`	Changes a task's status	✓
`log_decision(decision, by, time, reason)`	Adds a row to Notion Decisions	✓
`add_requirement(story, criteria, moscow)`	Adds a row to Notion Requirements	✓
`build_bom(system)`	Returns a part / qty / cost / supplier table	✓
`compare(option_a, option_b, criteria)`	Returns a trade-off table (e.g. 60 vs 75 kWh)	✓
`save_meeting()`	Writes summary + notes to Notion Meeting Notes, returns the link	✓

How it should answer

Rule	Expected
Chat replies ⩽ 6 lines or one small table	✓
Numbers always come with assumptions or a source doc	✓
Confirms every Notion write in chat ("✓ Task added: Ali — cost sheet — Fri")	✓
Asks one clarifying question when a request is unclear	✓
Interrupts the meeting on its own	✗
Invents specs, prices or suppliers	✗

Live test

3–4 TechSol staff discuss **Project Falcon** (30,000 USD electric SUV). Your agent must: post the consent notice, compare 60 vs 75 kWh packs, log the battery decision, create tasks from speech and on request, write user stories, list the main parts and suppliers for the 60 kWh pack with a cost estimate, recall the decision, fill Notion and post the link.

Resources

Provided	✓	Not allowed	✗
Meeting account (your platform)	✓	Real company data	✗
Notion workspace + 4 databases	✓	Production accounts	✗
Cloud GPU VM for deployment (24 GB GPU)	✓	Hosted bot / transcription services	✗
Fictional automotive pack: Falcon specs, parts list / BOM, supplier list	✓	Public chatbots on meeting data	✗
Private repo + point of contact	✓	Hard-coded answers	✗

Suggested free stack: Qwen2.5 7B / Llama 3.1 8B (Ollama), faster-whisper, Chroma or Qdrant. AI coding assistants allowed if declared.

Submit

Code + Docker + config · your two diagrams · Notion export · README with setup, tech choices and answers to: platform choice, latency budget, wake-word design, speaker tagging, stopping invented specs.